Haijian Sun

Assistant Professor of Electrical and Computer Engineering

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Research Interests

AI-Native Wireless and Learning Systems: Data-driven and physics-informed wireless channel modeling; generative waveform and neural transceiver design; learning-enabled PHY/MAC optimization for adaptive communication and sensing.

Wireless Intelligence and Edge Autonomy: Multimodal data fusion for perception and communication; edge AI for real-time decision-making; trustworthy and energy-efficient communication in large-scale IoT and cyber-physical systems.

Experimental Systems and Prototyping: Software-defined radio platforms, O-RAN testbeds, and AI-ready wireless datasets for validation and open research.

Education

Ph.D. **Utah State University**, Electrical and Computer Engineering — Logan, UT, USA, 2014 – 2019
- Thesis: *Spectral, Energy and Computation Efficiency in Future 5G Wireless Networks*
- Ph.D. Scholar of the Year, 2019
- Advisor: Rose Qingyang Hu

M.S. **Xidian University**, Electrical Engineering (Telecommunication) — Xi'an, China, 2011 – 2014

B.S. **Xidian University**, Electrical Engineering (Telecommunication) — Xi'an, China, 2007 – 2011

Experience

University of Georgia, Assistant Professor of Electrical and Computer Engineering — Athens, GA, 2022 – present, 4 years

University of Wisconsin-Whitewater, Assistant Professor of Computer Science — Whitewater, WI, 2019 – 2022, 3 years

Mitsubishi Electric Research Laboratories (MERL), Research Intern — Cambridge, MA, May 2018 – Aug 2018, 4 months

Honors and Awards

Emerging Scholar Research Award — 2026
Recognizes scholarly excellence among assistant professors.

Best Paper Award — 2024

Find a Rover Challenge Award — 2024

CISE Research Initiation Initiative (CRII) Award — 2022

Engineering Research Initiative (ERI) Award — 2022

Chancellor's Innovation Award for Computer Science 2022

Regent Scholar Award 2021
The UW System's faculty award for excellence in undergraduate research.

Ph.D. Scholar of the Year 2019

ESI Highly Cited Papers 2019
Top 1% wireless communication journal publications in 2018 and 2019.

National Scholarship 2012

Distinguished Scholarship 2012
Awarded to students in the top 5%.

Funded Research

- **Rethinking Wireless Channel Through the Lens of Radiance Field** — Lead PI, NSF CCSS; $480K total, $280K UGA share; with Virginia Tech (2025–2028).
- **Intelligent Reflecting Surface Assisted Physical Layer Security for Ultra-dense IoT Networks** — PI, NSF CISE Core Small; $600K total, $225K UGA share (2025–2028).
- **SpectraGuru: Building an Open-Source Ecosystem for Raman and Spectroscopic Research and Education** — Lead PI, NSF POSE Phase I; $300K (2025–2026).
- **Tribal & Rural Autonomous Vehicles for Equity, Livability and Safety (TRAVELS)** — Co-PI, U.S. Department of Transportation; $15M total, $3.6M UGA share (2025–2031).
- **Towards Spectrum- and Energy-Efficient Large-Scale IoT Communications** — Sole PI, NSF CISE CRII; $229,998 (2023–2026).
- **Find a Rover Challenge Award** — Sole PI, NSF AERPAW; $2,000 unrestricted gift (2024).
- **Toward mmWave Vehicular Communication: A Multisensor Multimodal Deep Data Fusion Approach** — Sole PI, NSF ECCS ERI; $199,955 (2022–2024).
- **28 GHz mmWave Dual Channel Device** — Sole PI, TMYTEK device donation valued at $20,000 (2022).
- **Multimodal Deep Sensor Fusion for Next-Generation Wireless Vehicle Communications** — Lead PI, UW System Regent Scholar Award; $50K total, $40K share (2021–2022).
- **Privacy Protection among Community-Dwelling Older Wisconsinites in the Age of IoT** — Co-PI, Thompson Center; $105,757 total, $35,252 share (2021–2022).
- **Prototype Multimodal Sensors and Wireless Networks for Landslide Monitoring** — Co-PI, UW System Regent Scholar Award; $50K total, $16,666 share (2020–2021).
- **Dual-Antenna RFID Reader Prototype for Vehicle Localization** — Lead PI, Tokef LLC; $112,051 total, $56,026 share (2020–2021).
- **UGA intramural funding** — E-Mobility Initiative ($56K), Pre-Seed Grant ($4K), IIPA precision food-supply-chain tracking ($37.5K), and IIPA smart-farming infrastructure ($245K total), 2022–2024.
- **UW–Whitewater intramural funding** — Research Catalyst Grant, $10K (2021).

Publication Record

- Author of 80+ peer-reviewed journal and conference papers, with 3,000+ citations and an h-index of 28 as reported in the July 2026 CV.
- Co-author of *5G and Beyond Wireless Communication Networks* (Wiley, 2024) and a chapter on non-orthogonal multiple access in the *Encyclopedia of Wireless Networks*.
- See the SunLab publications page for the website bibliography.
- See Google Scholar for the current publication and citation record.

Patents

- Anti-theft system based on power-line carrier communication and dynamic image detection, China Patent Application 201310421200.X, Publication 103501188A (2013).
- Wiring-free indoor Wi-Fi access system based on power-line carrier communications, China Patent Application 201310069145.2, Publication 103200700A (2013).

Student Advising

Graduate Students: Lihao Zhang, Hanwen Zhang, Paul Kudyba, Xiang Ma, and Tilak Basnet.

Undergraduate Students at UGA: John Song, Andrew Primiano, Austin Song, Nivedha Natarajan, Sterling Strohauer, and Wayne Lam.

Undergraduate Students at UW–Whitewater: Jacob Gasser, Riley Redfern, Shane Flynn, Kevin Marks, Dakota Vaughn, Christian Ayon, Jake Schnor, Colin Janaes, David Ascherl, Javin Herath, Brodie Peyer, JT Anderson, Bryce Erdman, Caitlin Seibel, Jeremy Angles, Brett Steck, Nick Huffman, and Gabriel Lewis.

K–12 Students: Ethan Zhu, Green Canyon High School (2019–2023; now at UC Berkeley).

Selected Student Achievements

- Paul Kudyba, Hanwen Zhang, and Lihao Zhang received competitive student travel grants for IEEE INFOCOM and ICC (2024).
- Paul Kudyba placed second and third in the NSF AERPAW Find a Rover Challenge and received an IEEE INFOCOM travel grant (2024).
- Ethan Zhu received the IEEE ISICN Best Paper Award (2024).
- Jacob Gasser and Shane Flynn received the UW–Whitewater Undergraduate Research Program Award (2020).
- Jake Schnor was a UW Regent Scholar undergraduate researcher (2021), and Tilak Basnet received the UW–Whitewater Graduate Research Award (2021).

Selected Presentations

- **Radio Radiance Field: The New Frontier of Wireless Channel Representation** — NextG Channel Model Alliance, IEEE ISICN tutorial, Ericsson Research, and Samsung Research USA (2025).
- **HawkRover: An Autonomous Vehicular Communication Testbed with Multi-sensor Fusion and Deep Learning** — UW–Stout AI Club seminar (2024).
- **Driving the Future: Wireless Communication Solutions for Electrified Transportation** — Asian American Engineer of the Year, New York (2023).
- **Approximate Wireless Communication for Federated Learning** — ACM WiSec Workshop (2023).
- **Efficient, Secure, and Intelligent Wireless Connectivity for Billions of Devices** — invited seminar series at six U.S. universities (2022).
- **Open Radio Access Network and Learning Algorithms for Next-Generation Massive MIMO Applications** — IEEE Future Networks seminar (2021).
- Additional invited and conference presentations include IEEE i-ETC, SPAWC, ICC, WCNC, Globecom, Qualcomm, Sony, and Utah State University (2015–2020).

Teaching

University of Georgia: ELEE 3270 Microelectronics I; ELEE 8930 Optimization Theory and Engineering Application; ELEE 8990 Advanced Topics in Wireless Communication.

University of Wisconsin–Whitewater: Taught 19 classes across six courses, developed three new courses, and advised capstone and thesis work in programming, systems, cybersecurity, forensics, and embedded-system security.

Utah State University: Graduate teaching assistant for Control Systems, Microelectronics I, Computer Networks, and Wireless and Mobile Networking; also served as a laboratory session assistant.

Professional Service

- Associate Editor, *IEEE Communications Magazine* (2024–present); Editor, *IEEE Network* (2023–present).
- NSF CISE Core program panelist (2024, 2025).
- Lead organizer, First IEEE Workshop on UAV Communication and Experimentation at IEEE VTC Fall 2024.
- Technical Program Committee Chair for IEEE WCSP 2023 and IEEE ISICN 2024.
- Technical Program Committee member for IEEE ICNC, ICC, and INFOCOM across multiple years.
- Completed more than 200 reviews for leading IEEE and ACM journals and conferences.
- Co-chair, IEEE Future Networks Massive MIMO group (2018–2021).
- University service includes UGA ECE graduate program, curriculum, and faculty search committees, as well as extensive program and committee leadership at UW–Whitewater.

Media Coverage

- NSF coverage of SunLab's AERPAW Find a Rover Challenge award.
- UW–Whitewater feature: *Hands-on Research with an Impact, Making Self-driving Cars Smarter*.
- Coverage by Spectrum News 1, *The Gazette*, the University of Wisconsin System, and the UW–Whitewater News Room for cybersecurity education, research, and the Regent Scholar Award.